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Policy-mix evaluation: Governance challenges from new place-based innovation policies

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ABSTRACT

The policy mix concept has become popular in innovation policy literature. It is particularly relevant for complex challenges such as sustainable industrial transitions that require joined-up interventions from different policy domains. Yet finding the right policy mix for a given challenge is strongly conditioned by the governance context in which individual policies emerge and evolve over time. In particular, the evaluation of policy mixes in a regional context is often neglected.

This paper aims to deepen understanding of the interaction between governance processes and policy mix evaluation in the specific context of smart specialisation strategies. These newly popular regional innovation strategies are ideal laboratories for analysing policy mixes due to their place-based nature, their directionality, their experimental character and potential orientation towards grand societal challenges, and their complex governance context.

The paper builds on the smart specialisation literature, the innovation policy and policy evaluation literature, and the literature on governance in pluralistic contexts to build a conceptual framework for analysing the governance of policy-mix evaluation. The conceptual arguments are illustrated by the Basque Country case in Spain, providing insights on the establishment of a strategic innovation policy mix evaluation process in a complex setting of institutions and actors.

1. Introduction

The policy mix concept has become popular in innovation policy literature as a result of new combinations of policy rationales and a broadening and deepening of innovation policy (Flanagan et al., 2011; Borrás, 2009). Yet evaluating innovation policy mixes in practice is a difficult task (Magro and Wilson, 2013). The difficulty arises more from coordination and governance issues than from specific technical issues. As such one of the main challenges is to look beyond impact evaluations that search for optimum policies and by extension for an optimum policy mix. An optimum policy mix is elusive precisely because it depends on the dynamic institutional context in which the policies are embedded. That is to say, the behaviour and interaction of different components of the policy mix are conditioned by the place-specific and multi-level governance contexts through which policies emerge and evolve over time.

The focus of this paper is on the evaluation challenges that are emerging in new place-based innovation policies. In practice these policies are being strongly shaped by regional smart specialisation strategies that seek to generate structural transformation in the economy, which in many regions include the task of fostering sustainable transitions. Evaluating these integrated territorial strategies for research and innovation poses additional challenges to those that already exist around innovation policy mix evaluation (Magro and Wilson, 2015). It implies not only evaluating the specific impacts of policy-mixes, but also assessing how those impacts are contributing to the process and direction of structural transformation sought by the strategy.[1] In particular, evaluating policy-mixes within a dynamic territorial strategy accentuates the importance of governance. This is because policy design and implementation become integrally related to broader decisions on the direction of the territorial strategy (e.g. which R&D activities to prioritise) at a local or regional level.

There is a great deal of literature devoted to exploring the concept of governance with relation to public policy (Bevir, 2012; Davoudi et al., 2008; Jessop, 1998; Lynn, 2012; Peters and Borrás, 2009; Pierre and Peters, 2000; Risse, 2012). For the purposes of this paper we are interested in the governance processes that characterise smart specialisation decision-making dynamics. This governance can be defined as

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[1] This is related to what Rogge and Reichardt (2016) refer to as policy-mix coherence; the extent to which policy-mixes are designed and implemented in a comprehensive manner.

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the dialogue, engagement and coordination through which decisions about strategic direction and accompanying innovation policy mix in a region are taken. Given the sensitivity of these types of strategic decisions, the interactions they entail between government and other regional actors with their own vested interests and the little experience of regional governments in prioritisation, new questions emerge around the roles that evaluation can play. In particular, there is potential to inject collective strategic intelligence into decision-making, a very specific need for regional governments with limited capabilities and capacities for prioritisation in an experimental context. Yet for such strategic intelligence to be considered robust and legitimate, the governance processes under-pinning the production and use of that intelligence warrant attention.

Governance of evaluation is often neglected in the evaluation literature. Debates have traditionally been focused on the external/internal evaluator issue with little emphasis on the process of governance itself. However, moving from individual policy evaluation to policy-mix evaluation highlights the importance of evaluation governance. Innovation policy mixes are normally designed and implemented by different government actors operating at different administrative levels (Magro et al., 2014). While some coordination in policy design and implementation is often achieved, a systemic coordination of evaluation is a more complex task, due in part to the sensitivity surrounding 'being evaluated'. In the context of smart specialisation strategies these challenges are particularly acute. There are multiple actors with a stake in research and innovation policy, both within government (different levels of governments impact regional strategies as do different departments and agencies), and outside government (in the other 'helixes' of business, research and civil society). These actors both engage in the governance process setting the direction of the strategy and are subjects and/or objects of evaluation of the policy-mix supporting the strategy. That raises potential conflicts of interest, as engagement in strategy shapes actors behaviour and that behaviour interacts with the policy-mix, which needs to be monitored and evaluated for policy learning purposes.[2] It is this link between behavioural change and governance of evaluation that remains unexplored in the literature.

The key contribution of this paper is to present a concrete view on the emerging governance needs of evaluation in the context of new place-based innovation strategies. It does so by dissecting characteristics of innovation policy mixes that are particularly important in the context of smart specialisation strategies; their verticality, their directionality and their capacity to foster experimentation. The paper then develops a conceptual framework for analysing the governance of policy-mix evaluation as an integral part of placed-based innovation policy. The framework is oriented to better understanding the conditions under which evaluation is able to provide collective strategic intelligence and contribute to the development of the government capabilities needed in smart specialisation strategies. These arguments are explored through the example of the Basque Country in Spain, which serves as an illustration of the different elements to be taken into account for establishing a collective, strategic evaluation process in a complex setting of institutions and actors.

2. Innovation policy mixes in the context of new place-based policies

Recent years have seen a general increase in attention afforded to the concept of policy-mix. This is evident in what has been called the third generation of policy-design theories (Howlett and Rayner, 2007, 2013; Howlett et al., 2015), in the innovation policy literature in general (Flanagan et al., 2011; Borrás and Edquist, 2013; Magro and Wilson, 2013; Lanahan and Feldman, 2015; Flanagan and Uyarra, 2016; Martin, 2016), and in literature on sustainable transitions specifically (Kivimaa and Kern, 2016; Kern et al., 2017; Rogge and Reichardt, 2016). This interest has been fairly rapidly passed onto policy-makers, at least at a superficial level, as can be seen in the increasing usage of the term policy-mix in policy documents. The European Commission's 2012 *Guide to Research and Innovation Strategies for Smart Specialisation* (Foray et al., 2012), for example, is explicit in establishing the 'definition of coherent policy mix' as one of six steps towards a smart specialisation strategy, and this has helped the terminology to become widespread among regional innovation policy-makers.

The analysis of policies seeking to foster sustainable transitions is an area where the significance of place is particularly evident, as seen in literature on the geography of sustainable transitions (Coenen and Truffer, 2012; Hansen and Coenen, 2015). Coenen et al. (2012: 968), for example, highlight the "institutional embeddedness of socio-technical development processes within specific territorial spaces", working alongside "an explicit multi-scalar conception of socio-technical trajectories". This is in line with Truffer's (2016: 12) identification of the need to disentangle complex multi-level geographical relationships so as "to better understand who plays a central role in the core processes and how policies may be defined that are able to intervene in these complex interaction patterns." Complexity of relationships between a wide range of different interested actors is indeed a common theme in analysis of place-based sustainable transitions (Hodson and Marvin, 2010, 2012; Loorbach and Rotmans, 2010; Maassen, 2012; Shove and Walker, 2010), and highlights the absolute importance of understanding the governance context in which strategy and policy decisions are made.

2.1. Smart specialisation and innovation policy-mixes

The place-based nature of sustainable transitions implies strong synergies with emerging theory and practice in the latest generation of regional innovation policies, so-called smart specialisation strategies. A coming together of science, technology and innovation (STI) policy with new industrial policy (Aranguren et al., 2017a; Radosevic, 2017a), these regional strategies provide a policy framework that follows the shift in literature and practice from spatially-blind to place-based innovation (Barca, 2009; McCann and Ortega-Argilés, 2015). Smart specialisation strategies are being implemented across Europe with impetus from the European Commission (including structural funding conditionality that insists all regions have a Research and Innovation Strategy for Smart Specialisation or RIS3). Rooted in previous place-based innovation policies, they shift the focus towards the explicit prioritization of activities based on regional capabilities. In many regions this prioritisation has been linked explicitly or implicitly to sustainable transition as a grand societal challenge (Capello and Kroll, 2016).

While in one sense smart specialisation strategies are an evolution of the regional innovation systems approach and acknowledgement of the desirability of a strongly place-based innovation policy, they have also been influenced by arguments for new industrial policies in which prioritization of activities is determined not by government, but by a nexus of private-public cooperation (Rodrik, 2004). Thus two distinguishing characteristics of smart specialisation strategies can be highlighted. Firstly, like industrial policies or mission-oriented STI policies, they are *directional*; they imply making choices. Secondly, they rely on *new governance processes* for making those choices; they are rooted in an experimental 'entrepreneurial discovery process' (EDP) involving a wide range of regional stakeholders from the quadruple helix of government, business, research and civil society (Foray, 2013, 2015).

It is the notion of an EDP that provides the key novelty of smart specialisation strategies. It implies that the governance process through which regional stakeholders identify the areas in which the region should specialise and build new paths becomes a central anchoring

[2] It is also worth noting that policy learning itself is usually not a shared purpose of all actors, so there will be asymmetries of interest.

point for innovation policies. As such innovation policies become necessarily rooted in evolutionary approaches. Although still very much a black box, how to build a sound EDP has been the object of increasing attention (Aranguren et al., 2018; Gianelle et al., 2016; Marinelli and Perianez Forte, 2017). Most critically for understanding policy mixes, an EDP implies a new governance configuration underpinning regional innovation strategy and policy. It transforms them into territorial, rather than government, strategies and policies. This new governance is a highly pluralistic context (Lynn, 2012; Howlett and Ramesh, 2014; Jarzabkowski and Fenton, 2006) in which building shared vision requires navigating through a range of different interests and potential conflicts among actors. As such, Morgan (2017: 569) has argued that it "places unprecedented demands on public sector bodies".

Another key feature of smart specialisation strategies is their treatment of research and innovation as transversal elements, influenced by and impacting on government policy from the whole spectrum of government departments. Developing a coherent policy mix, roadmaps and action plans to promote regional priorities is one of the six steps promoted by the European Commission (Foray et al., 2012). Yet policy-mixes in the smart specialisation context have tended to be considered more a portfolio of instruments from which to pick and choose, than a policy setting that might be strategically used to affect the directionality of innovation, fostering a structural transformation of the economy through experimentation. This is somewhat incongruent given the aim of smart specialisation strategies to generate structural transformation through targeting specific innovation activities over others.[3]

There is indeed little reference to new innovation policy instruments or new combinations of instruments in the smart specialisation literature, and Veugelers (2015) and McCann and Ortega-Argilés (2016) find relative homogeneity of policy mixes across Europe. Moreover, very little emphasis has been placed on the link between the government role in designing and implementing an appropriate policy-mix and the dynamic, pluralistic EDP process through which strategic direction should emerge. Directionality through experimentation, which is at the core of the EDP, tends to materialise in practice through the selection of projects to fund. How the dynamics of that project selection are linked to policy-mixes remains underexplored in two ways. The first is in terms of how prioritised projects are supported by policy-mixes. This relates to neoclassical failures, where policy can rely on classical incentive mechanisms. The second is in terms of how policy-mixes might themselves contribute to generating experimentation spaces. This relates to coordination failures, implying that policy-mixes should include 'softer' instruments (Rodríguez-Pose and Wilkie, 2017). In practice, however, the experimental approach to smart specialisation strategies has been confined to the selection phase, and implementation is recurring to traditional instruments (Radosevic, 2017b). This poses a particular policy-mix challenge given acknowledgement that the EDP should be continuous and present throughout the implementation of smart specialisation strategies (Marinelli and Perianez Forte, 2017).

2.2. Dissecting the innovation policy mix

There are several definitions of the policy-mix concept, some of them stressing the complex nature of the term (Flanagan et al., 2011), and others focused on the issue of which elements should be combined in a policy-mix (Borrás and Edquist, 2013). Rogge and Reichardt (2016) propose a holistic view of the term and highlight that a policy mix is not only made up of concrete elements (instruments), but also by processes (policy-making) and their characteristics (mainly consistency of elements, coherence of processes, credibility and comprehensiveness). Howlett et al. (2015) propose a typology of policy mixes ranging from simple cases in which there are single instruments from single policies and one single government, to more complex ones (multi goals, multi policies and multi-level governments). Regional innovation policy is in this respect considered a complex setting (Magro and Wilson, 2013), in which we can find different instruments from different policy domains that respond to different rationales, target different actors, and are designed and implemented over time by governments at different levels.[4]

However, one of the main criticisms to the regional innovation policy literature – including the European Commission's Smart Specialisation Guide (Foray et al., 2012) – is that it tends to list innovation policy instruments as options from a portfolio that policy-makers can choose from, contradicting the notion of 'one size doesn't fit all' (Todlting and Trippl, 2005). There is little explicit reflection about the rationales of each instrument, the policy goals they can be addressing, or the overall policy strategy they respond to. This is especially problematic given consensus in policy design literature that Tinbergen's rule of a one-to-one correspondence between instrument and policy goal no longer holds. Thus one instrument may address several targets or several instruments address the same target. Additionally, policy mixes are built over time and normally result from incremental changes that introduce modifications to existing instruments or add new ones to the previous mix (Flanagan et al., 2011; Streek and Thelen, 2005). This is reflected in the often little room for manoeuvre policy-makers have to introduce changes to historically-rooted practices (Howlett and Rayner, 2007).[5]

To further dissect the innovation policy mix concept in the context of smart specialisation strategies, it is particularly relevant to consider three dimensions in greater detail; their verticality, their directionality, and their capacity to facilitate experimentation.

2.3. The verticality of policy-mixes for smart specialisation

The multi-level reality for regions has strong implications for the consistency of policy-mixes. The absence of contradictions among policy objectives, the presence of synergies among instruments and the alignment of the instrument mix to the evolving regional innovation strategy become even more complicated questions when the administrative boundaries of policy-mixes are blurred. Thus the capacity of regional actors to generate coordination mechanisms between government levels becomes critical.

On the one hand contradictory objectives could come from different policy interests; e.g. national goals to champion certain activities by investing in large infrastructures versus regional goals to promote the activities coming from the local EDP. On the other hand the type of instruments integrated into the policy-mix is likely to depend on both the geographical scale at which policies are designed to impact and the level of government responsible. There are some instruments that are considered tools for national governments and have hence been neglected in the place-based policy literature, although they are included in the portfolio of possible instruments to be implemented by regional governments. The use of public procurement for boosting innovation is a good example of this (Uyarra et al., 2017). Network or cluster policy instruments, on the other hand, are most commonly implemented at regional level, due to their lower cost and their reliance on proximity relations.

[3] In this sense the rapid uptake of smart specialisation strategies provides a great example of the ubiquity of use of the policy mix term with little depth of understanding around what it could imply.

[4] The implication of different government levels equates to the notion of policy-mix 'verticality' posed by Howlett et al. (2015). This can be seen, for example, in practical application in Matti et al. (2017) analysis of the policy mix underpinning the development of the Spanish wind energy sector, or in the analysis of Bodas Freitas (this issue) or Huang (this issue).

[5] In the case of regional innovation policies, limited room for manoeuvre may also be due to lack of competences of regional governments over one or several of the instruments and/or lack of capabilities for implementing more radical changes.

2.4. The directionality of policy mixes for smart specialisation

Policy-mix directionality is an issue that has not been much considered in either the general literature on policy design or that on innovation policy. While the focus of the policy-mix concept tends to be put on maximising the complementarity between instruments (Howlett et al., 2015), how to address directionality is often forgotten. Indeed, European regions had a 'menu' of instruments in the RIS3 guide (Foray et al., 2012) according to different taxonomies, but there was no clear indication about the directional or neutral nature of the instruments to be implemented. Regions therefore face the challenge of building consistent policy-mixes as described by Rogge and Reichardt (2016) that explicitly target prioritised activities through experimentation.

Many regions have followed a 'patching' path (Howlett and Rayner, 2013), essentially modifying existing instruments in order to address new challenges or just keeping the traditional ones, which minimise the experimental nature of smart specialisation strategies (McCann and Ortega-Argilés, 2014). Although Howlett et al. (2015) suggest following this strategy in established and complex environments, it can be risky in the framework of smart specialisation strategies. In lieu of a more comprehensive re-design of the policy mix, there is a danger that patched policies may simply be supporting projects or activities labelled as 'prioritised' by actors in order not to lose previous support, leading to opportunistic behaviour. In addition, it is difficult to address certain specific problems faced in many smart specialisation strategies with this approach; for example the problem of reducing scaling-up problems in certain key enabling technologies, like nanotechnologies, that are important for the development of new innovation paths from other priority areas.

An alternative solution is to design tailored instruments for each priority and priority goal. Yet this generates high administration costs and may not yield high additionality. It is here therefore where real concern with policy-mix could add a lot to the strategic design of policies to support smart specialisation; for example through a policy-mix design for each priority area that combines neutral instruments for the whole economy with tailored instruments addressing specific failures. This would require effective coordination mechanisms to guarantee coherence. Moreover, the dynamism inherent in the EDP and the plurality of interests contained within that process makes it critical that there is strong strategic intelligence around how this policy-mix is working and evolving to meet new needs.

2.5. The challenge of experimentalism

Building on the above arguments it is clear that the practical challenge of policy-mix design and implementation in the context of regional smart specialisation strategies goes beyond improving previously existing mixes of instruments. Innovation policy mixes to support smart specialisation strategies should be composed of directional and horizontal instruments (from various domains and administered by different government levels) that promote specific innovation investments, but that also facilitate processes of experimentation (Magro and Wilson, 2013; Reid and Maroulis, 2017; Rodríguez-Pose and Wilkie, 2017). Yet smart specialisation literature frequently identifies as a major obstacle the lack of regional capabilities to develop an experiment-based regional policy (Foray, 2017; Marques and Morgan, 2018; Radosevic, 2017b).

The remainder of the paper explores how the strategic intelligence and learning provided by appropriately governed evaluation processes might be part of the solution to boosting regional capabilities, both with respect to experimental processes and with respect to effectively managing the aforementioned challenges of policy-mix verticality and directionality. A key argument is that new place-based innovation policies convert policy-mix evaluation from an operational to a strategic issue; it becomes a collective intelligence process that is central for avoiding coordination failures among different policy levels, for effectively navigating decisions around strategic direction, and for fostering meaningful experimentation. This in turn opens a series of questions around the governance of policy-mix evaluation.

3. Evaluation governance for smart specialisation policy-mixes

Despite an adoption of the policy-mix language within the innovation policy-making community, there has been little analysis of how the singular characteristics of policy-mixes for place-based innovation that are analysed above shape evaluation practices and *vice versa*. Monitoring and evaluation clearly has large potential to contribute to understanding the behaviour and performance of policy-mixes, going beyond accountability purposes to focus on learning. While the results of policy learning will differ depending on the object and subject of learning (Bennett and Howlett, 1992), the timely injection of monitoring and evaluation knowledge will be a key input. However, innovation policy-mix learning requires a shift from fragmented evaluation of the effectiveness of individual instruments (Martin, 2016) to more holistic evaluations accounting for interactions among instruments in a complex policy context (multi-rationales, multi-level, multi-actor) (Magro and Wilson, 2013).

In the context of smart specialisation strategies policy learning around the directionality of policy instruments should be a prime concern (Magro and Wilson, 2015), as should learning related to the effectiveness of experimentation processes and to vertical policy coordination. Yet while monitoring and evaluation is acknowledged as important for smart specialisation – featuring strongly in both of the Commission's guides (Foray et al., 2012; Gianelle et al., 2016) – the emphasis on how they can contribute to strengthen the critical link between implementation of the innovation policy-mix and the dynamic, experimental nature of an EDP is largely missing. In practice only very conservative approaches employing traditional indicators to monitor smart specialisation strategies tend to be found (McCann and Ortega-Argilés, 2016).

It is argued here that concern with the appropriate governance of policy-mix evaluation in highly pluralistic contexts is likely to be central to its usefulness as a strategic intelligence tool that can boost often-lacking regional capabilities and guide both the EDP and the dynamic design and implementation of the innovation policy-mix.[6] In particular, the new governance arrangements embodied in the notion of a territorial EDP should be extended to evaluation. This marks a significant change for an activity that is usually conceived as a government-centred (and relatively closed) process of gathering and analysing information; it requires "fourth generation" evaluation practice in Guba and Lincoln's (1989) classification.

3.1. Changing the paradigm: evaluation as social learning

Del Río (2014), pleading for a better definition of policy boundaries in order to assess the success of a given policy-mix distinguishes between horizontal (single-level) and vertical (multi-level) mixes. While the vertical dimension is the most complicated for assessing policy-mixes (Stame, 2004), the horizontal dimension also has potential to be a source of vested interests among different regional actors, particularly in the context of a pluralistic EDP. Tackling this complexity requires changes of evaluation paradigm that are only possible through social learning (Bennett and Howlett, 1992). Indeed, following Del Río (2014), and also Kulhman's (2001) arguments on vested interests (winners at a superior administrative levels might lead to losers at inferior levels), it can be argued that a framework for evaluating policy-mixes in the context of smart specialisation strategies should be

[6] This is in line with the arguments of McCann and Ortega-Argilés (2016) around the importance of learning by evaluating in place-based policy experiments, and reflects the emphasis on experimentation in much of the literature on sustainable transitions.

explicitly aware of these (conflicts of) interests and should explicitly seek to overcome them through fostering processes of social learning.

Such an approach might find conflict with the European Commission's primary concern with 'accountability evaluation' of the funds invested in smart specialisation strategies, leading to un-useful evaluation practices (Amanatidou et al., 2014). Moving beyond this dichotomy requires working on the complementarities of the different approaches and appreciating how evaluation evidence of different kinds, when embedded in appropriate governance arrangements (Howlett and Rayner, 2013), can generate collective, strategic learning. In particular, it implies recognising the significance of behavioural additionality in terms of promoting experimentation and uncovering appropriate strategic direction, alongside the more traditional evaluation measures of input and output additionality.

As such a better understanding of innovation policy-mixes for place-based policies doesn't only depend on finding appropriate evaluation methods, but also on how evaluation is governed. Recent literature on monitoring and evaluation of smart specialisation strategies puts emphasis on governance structures such as evaluation committees as a passive mechanism for evaluating progress (see Gianelle and Kleinbrinck, 2015). It doesn't consider how governance processes interact actively with the evaluation of smart specialisation strategies and their policy-mixes, shaping actors behaviour. Yet literature on evaluation emphasises not only on the formative role of evaluation for policy learning, but also the importance of people for understanding change (Pawson and Tilley, 1997) and for making evaluation useful (Patton, 2012). Indeed, the principles of empowerment evaluation, namely training, facilitation and advocacy (Fetterman, 1994), integrated into emerging smart specialisation governance arrangements, could constitute a powerful source of change and contribute to promote social learning capabilities.

3.2. Towards a conceptual framework for policy-mix evaluation governance

Among the dimensions that make innovation policy-mixes quite singular in the context of smart specialisation strategies and related sustainable transitions, the analysis of the previous section highlighted verticality and directionality in an experimentalist context. In this extreme pluralistic context (Gil and Pinto, 2017) the vested interests of multiple actors at different levels are likely to be present, and a sophisticated network governance solution will be required to navigate these interests. This governance situation will also condition the evaluation of policy-mixes, leading to different answers to the questions of 'why evaluate?', 'what to evaluate?', 'how to evaluate?' and 'who should be responsible for evaluation?'. Indeed, the collective results and usefulness of evaluation will depend on the forces of the different private and public actors embedded in the EDP and with vested interests in the evaluation outcomes.

In less complex innovation policy situations, the main purpose of evaluation is likely to be accountability and control, as policy decision-making is principally within-government. This may lead to fairly traditional evaluation processes (the first and second generations of evaluation practices following Guba and Lincoln, 1989), which rely on either internal or external evaluators conducting evaluations based on the impacts and cost effectiveness of the mix of instruments being employed (including their interactions). However, the plurality of agents involved in the policy process surrounding smart specialisation strategies alongside the particular sensitivity of directionality, where conflicts in interest are likely, changes the requirements. Given the sensitivity and subjectivity of certain evaluation information – e.g. around the 'success' of following certain innovation pathways over others – it will be very important that all parties buy into the evaluation process and that it is governed in a way that is seen as legitimate. Only under those circumstances is evaluation likely to be accepted as impartial strategic intelligence that can legitimately inform specific decisions on both the policy-mix and the overall strategic process in which agents are engaged through the EDP.

This assumption does not imply a rejection of more traditional evaluations for individual instruments with a purpose of accountability (for example evaluation of smart specialisation strategies to fulfil the European Commission's requirements), however, but rather an integration of those into systems evaluations in which other evaluations are more learning-oriented (Arnold, 2004). Nevertheless, evaluation of innovation policy-mixes in the context of smart specialisation strategies should move from being primarily about accountability to being a strategic learning tool at different levels. This implies a focus on learning about the contribution of policy-mixes to the directionality of the strategy, but also the capacity to detect changes in actors' behaviour towards a more experimental or entrepreneurial character, in essence evaluating the EDP itself.

Approaches like stakeholder-evaluation (Hansen and Vendung, 2010) are more suitable for such governance contexts than other traditional evaluation approaches. Yet, the main unresolved issue does not rely on the specific methods to be used in discrete evaluation tasks but on the overall governance of the process. That means not only involving stakeholders in the evaluation, but turning it around to put them at the core of the process, following principles of empowerment evaluation (Fetterman, 1994). The 'owners' of the policy instruments that compose the policy-mix, the beneficiaries of those instruments and other related stakeholders essentially shift from being object of evaluation to protagonists. The objective is to facilitate collective strategic intelligence and learning that leads to changes that improve decision-making and experimentation processes related to the smart specialisation strategy.

These ideas are illustrated in Fig. 1, which depicts three possible governance modes for the evaluation of innovation policy-mixes. These move through different scenarios of policy-mix complexity, which are argued to shape the governance requirements of policy-mix evaluation. The last scenario corresponds with the characteristics of smart specialisation strategies – in particular the significance of directionality, verticality and the need for experimentation. It is important to note, however, that the three types of evaluation and their associated governance modes could co-exist – and indeed are likely to co-exist – in the framework of a systemic evaluation (Arnold, 2004) that brings together different pieces of evidence in a strategic learning framework.

By way of summary, Table 1 sets out the evaluation governance challenges that emerge in the preceding discussion as they correspond to each of the key characteristics of innovation policy-mixes supporting smart specialisation strategies. In the next section we build on the arguments developed so far by using the Basque Country example to illustrate the governance challenges involved in designing an evaluation framework for innovation policy-mix alongside the implementation of a highly pluralistic EDP.

4. The Basque country smart specialisation strategy: an illustration of policy mix evaluation governance challenges

4.1. Background to the case

The Basque Country, a region of two million people in the north of Spain, has been object of analysis for many innovation scholars in recent years due to what is usually seen as successful industrial transformation and innovation upgrading. It is one of the few European regions considered as a 'real' regional innovation system (Cooke et al., 2000) and one of the few that already had an incipient smart specialisation strategy even before the term was conceived (Aranguren et al., 2016). In addition, the Basque Country holds strong regional competences on innovation policy which exist alongside limited national competences and significant sub-regional competences. Thus the region can be defined as a policy space in which innovation policies from different levels are being felt (Uyarra and Flanagan, 2010), as a consequence of which innovation policy and its derived instruments are embedded in a complex, multi-level policy framework.

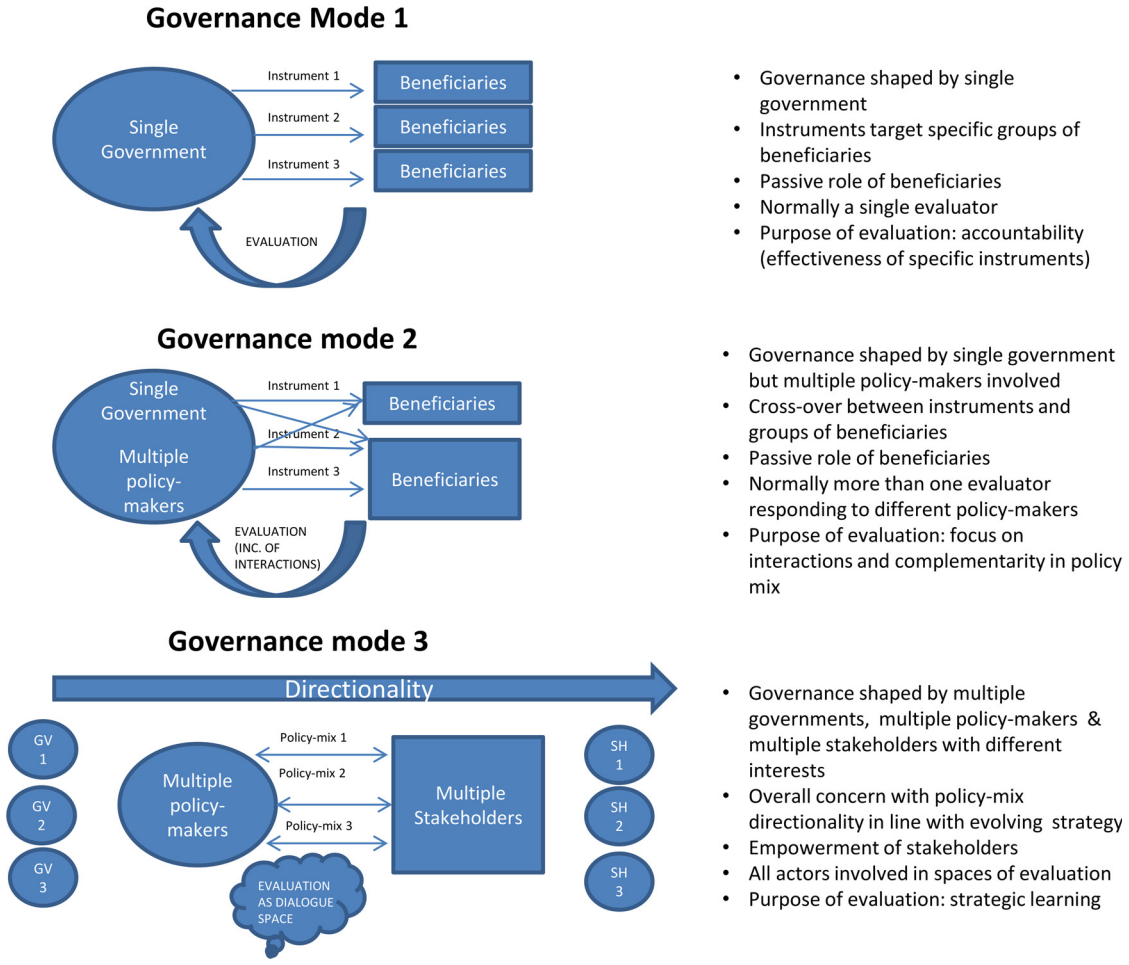


Fig. 1. Three governance modes for policy-mix evaluation.

Another key characteristic of regional innovation policy in the Basque case is strong continuity and cross-party consensus. Indeed, since the Spanish autonomous communities were created at the end of the 1970s the Basque region has always concentrated resources on reinforcing and improving industry as the basis for its competitiveness, which has led to a strongly-resourced innovation policy. This path has also led to the development of strong competences and capabilities among regional policy-makers (Walendoski et al., 2011). However, monitoring and evaluation has been identified as a weak link in Basque innovation policy, as highlighted by the peer-reviews conducted by OECD (2011) and Morgan (2013).

This section builds primarily on analysis of secondary documentary evidence. However the authors also use 'grey data', from more than 15 informal interviews that have been conducted with regional policy-makers and stakeholders and from processes where they have worked alongside relevant stakeholders (see Aranguren et al., 2016 and Aranguren et al., 2017b).

4.2. Policy mix in the framework of the Basque smart specialisation strategy

The smart specialisation strategy of the Basque region is set out in the latest Science, Technology and Innovation Plan (PCTI 2020). This reflects the strategic character of the innovation policy-mix. Although many of the pieces of the current smart specialisation strategy were previously in place, the European Commission's agenda was taken as an opportunity to build on and improve existing plans. The current strategy is focused on three main priority areas and four opportunity niches, each of which was selected on the basis of existing science and

Table 1
A summary of challenges for policy-mix evaluation governance.

Characteristics of policy-mixes supporting smart specialisation	Challenges related to evaluation governance
Combination of directional and neutral instruments	Deal with plurality of vested interests (and existence of policy silos)
Multi-level or vertical dimension	Deal with coordination of evaluation information across levels of government
Promoting experimentation	Able to detect and integrate into evaluation practices impacts on actors behaviour and structural systemic changes in the context of a dynamic EDP
Multiple actors as both policy-makers and beneficiaries	Able to support government and other actors in developing the capabilities required to adapt the policy-mix to the evolving strategy

technology and/or business capabilities in the region.[7] Among these priority areas, advanced manufacturing stands out both in terms of its importance to the regional economy and the pluralism of interests involved, and constitutes a good focal point for our illustration:

> *"Advanced Manufacturing is a complex environment, even in the Basque Country, as plenty of agents need to be taken into account, heard and followed. It's a multisector priority, with automotive industry, train and ship manufacturers, metal industry, aeronautics, machine-tool manufacturers, even ICT - all of them are involved. Each industrial sector has its own historical and operational dynamics, so joining visions, steps and strategies requires a really intensive-cooperation driven process. Even in the public area, and although the Department of Economic Development and Competitiveness is leading the process, there are clear links to other departments, institutions…"* (Aranguren et al., 2016 p. 46).

This complex environment is reflected in both the governance of the EDP surrounding this priority (see later analysis), and in the specific policy-mix for addressing the priority, which are mainly instruments from the regional government, although instruments from different administrative levels are also included. They are principally designed and implemented by two of the main regional government departments: the Department of Economic Development and Competitiveness (DEDC) and the Department of Education.

Other innovation policy instruments at sub-regional levels that impact in the region include those implemented by the Provincial Councils of Bizkaia and Gipuzkoa, among which we can highlight tax incentives for R&D. In addition, there are city-level economic development initiatives, for example from Bilbao city council, which are mainly related to the use of urban spaces and encouraging cross-fertilisation across activities. At the supra-regional level, also, there are some national and European STI policies (for example the H2020 programme) that impact on firms and knowledge infrastructures related to advanced manufacturing in the region. Thus there is a strong *verticality* to the Basque innovation policy mix for advanced manufacturing, with at least five different administrative levels involved.

Analysis of the *directionality* of the policy mix is less straightforward. While the smart specialisation strategy by definition introduces directionality into the strategy for innovation, it can be argued that the Basque policy-mix for advanced manufacturing is composed of a mix of instruments that exhibit different degrees of directionality. There are some neutral instruments with no explicit directionality, such as those that support non-technological innovation in firms. There are also instruments that are directional *per se*, such as those that support scientific and technological agents focused on specific domains and technologies. Finally there are instruments that have been 'made' directional to serve the strategy after a patching process.

This last group warrants further explanation. The Basque Country has always been characterised by a high degree of support to industrial firms based largely on subsidies. One of the characteristics of these support mechanisms has been their lack of directionality. However, in recent years and especially as a consequence of the implementation of the smart specialisation strategy, the Basque Government has introduced as criteria for eligibility to subsidized projects that they have to be conducted in one of the 'prioritised domains' of the strategy. In that sense, advanced manufacturing projects will be prioritised over projects that fall outside of this domain. Although the instrument is not directional by definition, the introduction of these criteria in a patching process has introduced directionality.

4.3. New governance arrangements for smart specialisation

The degree of verticality and directionality of the policy mix in this case pleads for a new governance model based on experimentation in which the actors embedded in the EDP play a bigger role in the governance of the innovation policy mix and its evaluation. The response of the Basque smart specialisation strategy in terms of generating new governance arrangements can be split into two: 'hard governance' arrangements in terms of the regional government's oversight of the implementation and evolution of the strategy (and associated policy mix); and 'soft governance' arrangements in terms of the wider engagement, alongside government, of business, research and civil society in a so-called 'entrepreneurial discovery process'. Both respond in different ways to the network governance challenges posed by innovation policy-mix verticality, directionality and experimentation in a smart specialisation context.

The key novelty in the 'hard governance' arrangements or governance structures has been the establishment of an interdepartmental committee to coordinate the implementation of the strategy across the different government departments implicated and also to act as a focal point for vertical, inter-institutional coordination (Aranguren et al., 2016). This committee is composed of representatives of seven regional departments alongside the Presidency department, and its main role is setting milestones and supervising the strategy process by providing a meeting point to share what is happening in the development of the different priority areas. It is in this space where a shared vision is being built within government, an aspect that is particularly important for policy-mix purposes as it includes representatives from the different 'owners' of the instruments. The potential for improved communication and coordination between government departments around the innovation policy mix is recognised by those involved, and the discussions here also form the basis for stronger coordination with government institutions at different levels through an inter-institutional committee that meets after each inter-departmental meeting (Aranguren et al., 2016). Therefore, it could be argued that these new 'within-government' governance arrangements in response to the demands of the smart specialisation strategy governance favour better policy-mix governance in the region.

In terms of the 'soft governance' arrangements, the key novelty is the establishment of so-called 'live processes for the development of the priority areas'. In practice these are organised through 'steering groups' as spaces that bring together the key public and private agents, and form a focal point for the establishment of relevant thematic working groups to explore and develop each of the priority areas. In the case of the advanced manufacturing priority, for example, the composition of the steering group includes representatives from the Basque Government (Department of Economic Development and Competitiveness-DEDC and its agency-SPRI), from cluster associations and firms (automotive, machine tools, energy and ICT clusters), from knowledge and technology agents (technology centres and universities) and from intermediate agents.

In theory these soft governance arrangements should provide directionality and discovery of new activities based on experimentation over time, by identifying and exploring opportunities based on the competences and knowledge of the different agents involved. Government involvement in the steering groups and their reporting back to the interdepartmental committee should enable this directionality to feed into the innovation policy mix. In the case of the advanced manufacturing priority, almost all the regional actors with vested interests in developing an advanced manufacturing strategy are involved.[8] However, there is no direct representation from some of the key public agents that design and implement instruments included in the policy-mix, such as the regional department of education or the sub-regional representatives. Effective transmission of the emerging directionality to the different departments and levels of government involved in implementing relevant policy instruments thus depends on

[7] See Aranguren et al. (2016) for more detail.

[8] Indeed, there is an argument that the presence of the most dominant actors crowds out the involvement of small companies or entrepreneurs.

the connection between these soft governance arrangements and the hard governance arrangements of the interdepartmental and inter-institutional committees.

4.4. Evaluation context and challenges

Monitoring and evaluation has traditionally been considered one of the weaknesses of the Basque innovation system and policies (OECD (2011); Morgan, 2013). In recent years, however, the region has sought to push forward in this area and advances have been made in the evaluation framework established around the smart specialisation strategy. Evaluation of the smart specialisation strategy is conducted by Innobasque, the regional innovation agency, which has established a framework for evaluation that includes three different lines:

1 Monitoring the current strategic objectives and their evolution;
2 Assessing the contribution of individual instruments to the strategy (an exercise based on the monitoring system implemented by SPRI, the Basque Business Development Agency);
3 A comparative analysis of the Basque innovation system.

While this evaluation framework is in the early stages of implementation, by way of illustration it is interesting to assess it against the arguments made in the previous section. There we argued that innovation policy-mix in the context of smart specialisation strategies places different requirements on evaluation processes given the plurality of agents (public and private) involved in the policy process and the sensitivity of the directionality issue. More concretely, we suggested that the evaluation process itself should be governed in a way that is seen by all parties as being valid and leading to impartial strategic intelligence. This would imply moving from traditional evaluation methods to others such as stakeholder-driven evaluation, in which empowerment principles are present. In the context of this case, such a process could be an important tool for ensuring that the link between the soft governance and hard governance analysed above are effective and would feed the strategy with a learning loop.

While the monitoring of current strategic objectives and the comparative analysis of the Basque innovation system (objectives 1 and 3) are useful to understand where the region is in terms of overall direction and benchmarking with others, only the second objective of the framework can truly be termed a policy evaluation exercise. Moreover, it is fairly limited in terms of analysing the policy-mix as it is currently being conceived over one specific subsidy programme designed at the regional level. This implies that it will not easily be able to address verticality and directionality issues.

Indeed, the evaluation framework has been established for the strategy as a whole, as a Science, Technology and Innovation Policy, but there is not yet a consensus-built framework for each of the specific priorities, such as advanced manufacturing. Such a general approach has some disadvantages. First of all, evaluation will provide insights about general targets in the STI policy context (for example about R&D investments and their impacts), but little specific knowledge about how the policy is contributing to advance in each of the priorities and how the policy-mix contributes to the generation of experimentation and directionality in those areas. Secondly, the governance framework for evaluation will not take into account the specificities of each priority setting, missing an opportunity to take advantage of the dialogue spaces that are being established as part of the EDP to engage the different stakeholders in evaluation as a strategic intelligence tool to their discovery processes.

In the specific case of advanced manufacturing, for example, the framework takes into account only one of the government departments and levels responsible for relevant policy-mix instruments (the DEDC), and users (cluster associations, firms, knowledge organisations,…) are the object of the evaluation but not taking part in the exercise as part of their engagement in the RIS3. Following our arguments above, with this approach accountability purposes might be filled, but learning ones might be limited, inhibiting the strategic intelligence contribution of the evaluation to the regional smart specialisation strategy in general and to the articulation of a smart policy-mix in particular. There is, however, potential to promote consistency in the innovation policy-mix if evaluation governance evolves towards a new paradigm in which the different agents embedded in the EDP of advanced manufacturing are empowered in a more stakeholder-centric evaluation approach. Questions such as the coherence of the instruments with their goals and the strategy are already being discussed in a non-systematic way in the new forums created by the strategy. The inclusion of these issues in a more integrated evaluation mechanism would add legitimacy to the process, and contribute to the consistency and comprehensiveness of the policy-mix among the different regional actors.

Summarising, governance of smart specialisation strategies in the Basque Country has evolved towards a new governance mode (see figure 3) but evaluation processes still follow governance mode 1 or 2.[9] Evolving towards a governance mode 3 for evaluation in a way that complements other modes of governance will be a key step for generating collective learning about the contribution of policy-mixes to the evolution of the different strategic priorities.

5. Conclusions and emerging challenges

The notion of policy-mix has become very popular among innovation policy analysts and also among innovation policy-makers, where it has been given a boost by the rapid emergence of smart specialisation strategies as a new regional innovation policy framework. The key features of smart specialisation strategies make them extremely interesting laboratories for analysing policy mixes, in particular those geared towards promoting sustainable transitions Indeed, it can be argued that innovation policy under smart specialisation strategies, as with sustainable transition promotion, takes place in an extreme pluralistic governance context. Moreover, the emphasis in smart specialisation strategies on prioritization of activities as a result of an experimentation process (entrepreneurial discovery process) among private and public actors adds complexity in terms of the need for directionality of policy instruments in combination with neutral policy instruments and the active role of non-government actors in policy-making.

This new context for innovation policy-mixes creates demands for new governance arrangements and processes both within and beyond government. In particular, there is a need to ensure a well-functioning link between the different government departments and levels that design and implement an appropriate innovation policy-mix and the dynamic, pluralistic EDP process through which regional innovation priorities should emerge. These new governance arrangements in turn place demands on evaluation processes and in particular on the governance of evaluation for policy-learning purposes, meaning shifting from an accountability role of evaluation to a formative, strategic one. Indeed, where such a range of vested (and potentially conflicting) interests come together, it is important to ensure strategic intelligence through evaluation that is trusted and seen as legitimate by all parties. Thus the verticality and directionality of policy-mixes in placed-based policies with an experimental approach, condition not only evaluation itself, as new evaluation frameworks should be established in order to respond to the mix sophistication, but also the governance of evaluation processes. In particular, to complement traditional evaluation approaches a focus on strategic learning is required, alongside the integration of evaluation issues into the new governance arrangements underscoring entrepreneurial discovery processes and their links into government policy decision-making.

[9] The government is advancing towards a policy-mix evaluation and for that purpose is participating in an Interreg project called MANUMIX (www.interregeurope.eu/manumix/)

However, despite the smart specialisation agenda, the governance of policy-mixes in general and their implications for evaluation governance in particular, are still pending issues in regional-policy making. Even in regions that have developed capabilities in innovation policy-making over time, such as the Basque Country, incipient changes in policy-mix governance are not yet reflected in evaluation governance. Indeed, it is often naively assumed that once priorities are set through entrepreneurial discovery processes, governments will be able to choose from a menu of policies to guide innovation in those directions. Thus new governance arrangements capable of truly linking the strategic and policy processes are not adequately developed. These arrangements would add consistency to policy-mixes in the framework of smart specialisation strategies. Stakeholder-based evaluation and principles of empowerment evaluation could be employed as strategic tools in this context, increasing policy knowledge (and boosting its legitimacy) among the pluralistic collective of actors that play a role in the entrepreneurial discovery processes shaping innovation policy mixes.

The illustration presented here shows a work in progress in this regard. New governance arrangements relevant for the policy mix have emerged in the context of the smart specialisation strategy – both 'hard governance' mechanisms within government, and 'soft governance' mechanisms beyond government – but there remains a missing link in terms of a sound and comprehensive approach for strategic intelligence through evaluation integrated within these new governance arrangements. While this is one regional example employed for illustrative purposes, we argue that it is symptomatic of a wider problem, where there is still a governance gap in new place-based innovation policies between policy-mix design and implementation and policy-mix evaluation. Thus the linking of governance structures and processes for evaluation is a key issue in the context of place-based policy mixes. It suggests the need for further research into this governance gap drawing on what is being experimented with in practice in different regions as the implementation of their smart specialisation strategies takes form.

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